

Memory

Non-volatile
ROM

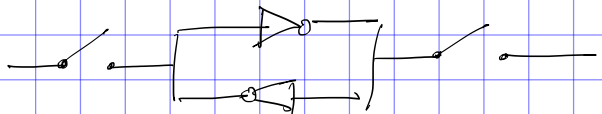
Volatile
DFF/flop
SRAM
DRAM

The core of volatile memory is either a capacitor holding the state or a back-to-back connected inverter.

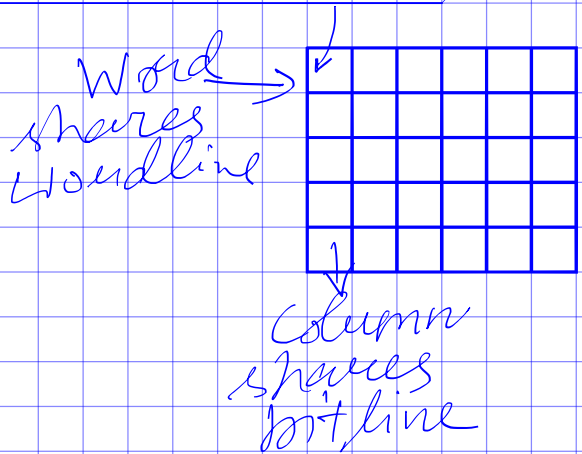
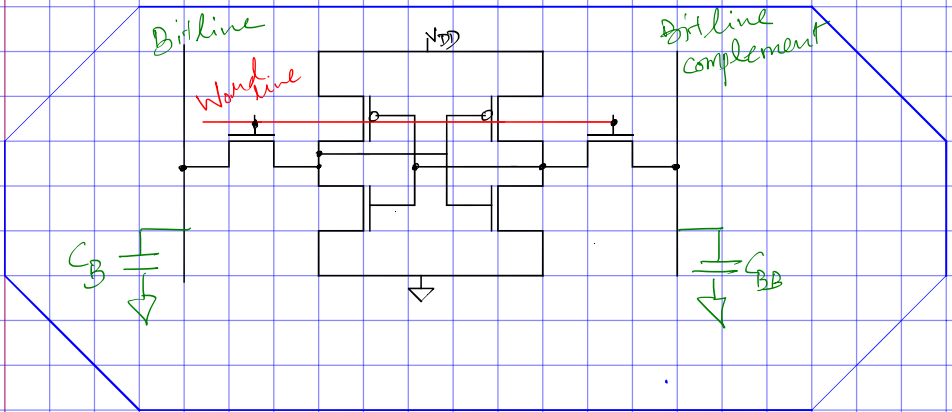
SRAM - Static Random Access Memory

Used to build arrays of storage often used in "fast" memory such as data/instruction cache of CPU. Each "bit" of memory has to be replicated millions of times typically depending on the memory hierarchy. This imposes the requirement of a 1-bit memory (called SRAM cell) to be as small as possible while providing reliable read/write/store operations.

Simplest implementation has 6 transistors and is called 6T SRAM cell. It comprises two inverters and two nMOS acting as switches.



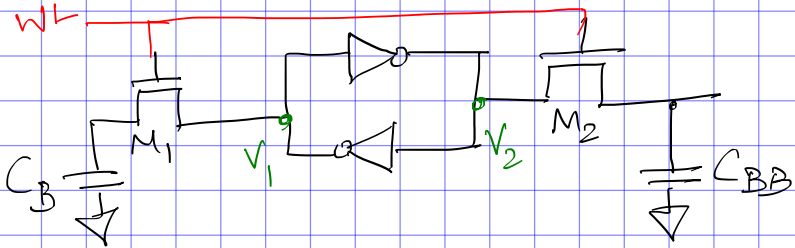
Cells are placed on a two dimensional grid. In a column, cells share two bitline capacitors. In a row, they share a wordline that enables the switches for read/write operation.



Assuming M rows and N columns a row decoder is used to select one of the M rows. M rows requires $(\log_2 M)$ address lines forms the select lines of the decoder.

A subset of B -bits among the N -bits in a row is chosen using a column decoder.
 $\# \text{ address lines} = \log_2 \left(\frac{N}{B} \right)$

Read operation



Steps:

- 1.) Precharge: Wordline WL is set to 0 and C_B & C_{BB} are charged to V_{DD}

- 2.) WL goes high:

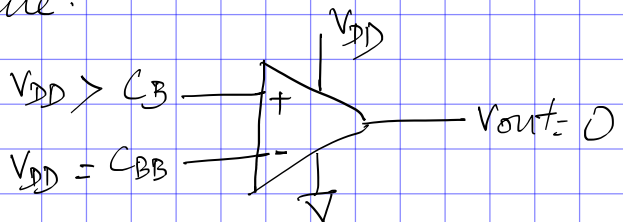
Let V_1 be the stored value & V_2 is its complement.

Assume $V_1 = 0$ (stored value is 0)

$$\Rightarrow V_2 = V_{DD}$$

Once WL goes high, C_B discharges to pull V_1 up. C_{BB} is held at V_{DD} since source & drain of M_2 are at V_{DD} .

A sense amplifier detects the difference between C_B and C_{BB} to output the stored value.

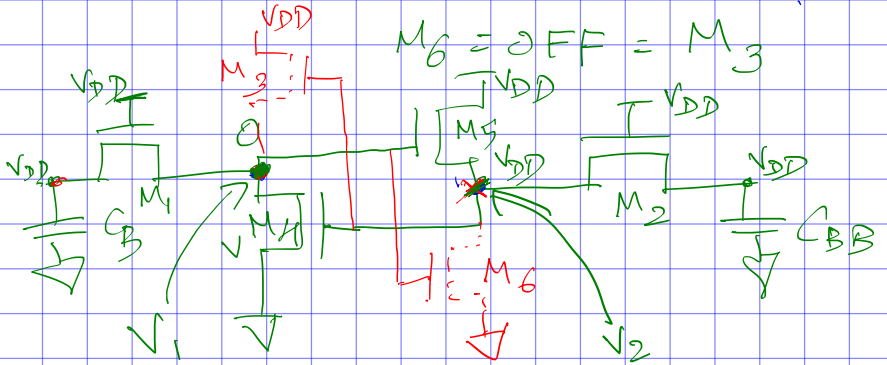


If the stored value is "1", $V_1 = V_{DD}$, $V_2 = 0$

$\Rightarrow C_B$ is held at V_{DD} and C_{BB} discharges the sense amp detects the difference.

3.) WL is set to 0 to stop V_1 or V_2 being pulled up.

Sizing transistors for read operation:

$$M_6 = M_3 = \text{OFT}$$


Once $WL = V_{DD}$, C_B discharges & pulls V_1 up. We do not want V_1 to be sufficiently pulled up turning on M_1 which could potentially flip the stored bit. $\Rightarrow V_1 < V_{Tn}$

When $V_i = V_{Tn}$

$$V_{DS}(M_1) = V_{DD} - V_{TN}, \quad V_{GS}(M_1) = V_{DD} - V_{TN}$$

$\Rightarrow M_1$ is in Sat. region

$$V_{DS}(M_4) = V_{TN}, V_{GS}(M_4) = V_{DD} \Rightarrow M_4 \text{ is in linear region.}$$

$$I_D(M_1) = I_D(M_4)$$

$$\frac{\mu_n C_{ox} \left(\frac{W}{L} \right)}{2} \left(V_{DD} - V_1 - V_{Tn} \right)^2 \times \frac{E_{cn} L}{E_{cn} L + V_{DD} - V_1 - V_{Tn}}$$

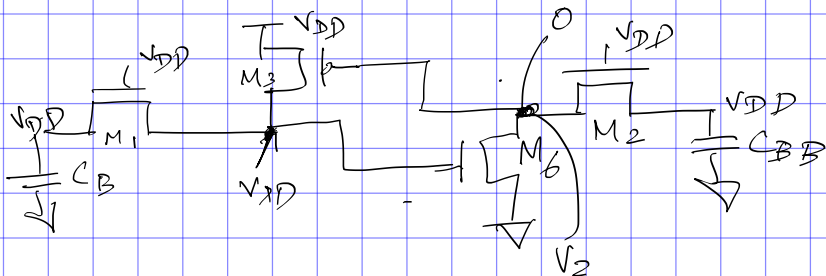
$$= \frac{\mu_n C_{ox} \left(\frac{W}{L} \right)}{2} \left(2(V_{DD} - V_{Tn}) V_1 - V_1^2 \right) \times \frac{E_{cn} L}{E_{cn} L + V_1}$$

$$\rightarrow \frac{W_1}{W_A} = \underbrace{\left(\frac{2V_{DD}V_{Tn} - 3V_{Tn}^2}{(V_{DD} - 2V_{Tn})^2} \right)}_{K_1} \times \frac{E_{cn} L + V_{DD} - 2V_{Tn}}{E_{cn} L + V_{Tn}}$$

If $K_1 \geq 1$, $W_1 = W_{min}$, $W_2 = K_1 \times W_{min}$

If $K_1 < 1$, $W_A \geq W_{min}$, $W_1 = \frac{W_{min}}{K_1}$

With the same argument M_2 & M_6 can be sized similarly using the below circuit.



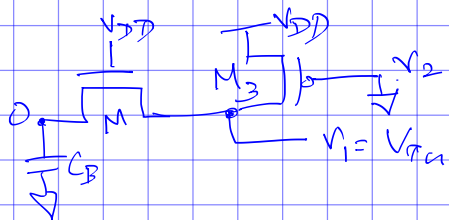
Once the switches M_1 & M_2 are turned off by setting $W_L = 0$, the inverters take care of pulling V_1 & V_2 to the right voltages.

Write operation

For M_1 to turn "ON" V_2 has to cross V_{Tn} which doesn't happen because of the way M_2 & M_6 are sized for the read operation.

This implies V_1 has to discharge to turn "OFF" M_6 and turn ON M_5 . This requires the effective resistance of $M_3 \gg M_1$, so that $V_1 \leq V_{Tn}$.

When $V_1 = V_{Tn}$, the circuit is as follows



$V_{DS}(M_1) = V_1 = V_{Tn}$, $V_{GS}(M_1) = V_{DD} \Rightarrow M_1$ is in linear

$V_{SD}(M_3) = V_{DD} - V_{Tn}$, $V_{SG}(M_3) = V_{DD} \Rightarrow M_3$ is in Sat.

$$I_D(M_1) = I_D(M_3)$$

$$\Rightarrow \frac{\mu_n C_{ox}}{2} \left(\frac{W}{L} \right)_1 \left(2(V_{DD} - V_{Tn})V_1 - V_1^2 \right) \frac{E_{cn} L}{E_{cn} L + V_1} = \frac{\mu_p C_{ox}}{2} \left(\frac{W}{L} \right)_3 \left(V_{DD} - V_{Tp} \right)^2 \frac{E_{cp} L}{E_{cp} L + V_{DD} - V_{Tp}}$$

$$\frac{W_1}{W_3} = \frac{(V_{DD} - |V_{Tp}|)^2 \times E_{cp} L \times \mu_p \times (E_{cn} L + V_{Tn})}{\underbrace{V_{Tn} (2V_{DD} - 3V_{Tn}) \times \mu_n (E_{cp} L + V_{DD} - |V_{Tp}|)}_{K_2}}$$

$$\boxed{W_1 \geq \frac{W_3}{K_2}}$$

Using the value of W_1 obtained from the read sizing size W_3 .

Similar argument applies for sizing M_2 & M_5 for writing "1".